

DEPARTMENT OF BUSINESS ADMINISTRATION
FACULTY OF ADMINISTRATION
OBAFEMI AWOLOWO UNIVERSITY, ILE-IFE, NIGERIA
BUS 304: ANALYSIS FOR BUSINESS DECISIONS
DON JAMES MONDAY (Ph.D., FNIM)

ASSIGNMENT MODELS

Assignment model is a mathematical technique that involves determining the most efficient allocation of people to projects, salesmen to territories, contracts to bidders, jobs to machines, etc. The essential feature of the assignment model is that each worker/machine must assigned one and only one task/job, so as to minimize project execution cost.

Assignment model helps project managers to make effective decision in allocating a task to a worker or machine that has the ability to execute the task efficiently; thus, enhancing productivity and profitability.

Problem Solution

Basically, there are TWO types of problem solution to an Assignment Model. They are:

- i) Enumeration method:** This method involves the listing of all possible combinations of assigning one job to one worker or machine. Enumeration is tedious for assignment problems involving more than 4 workers or machines.
- ii) Hungarian/Opportunity cost method:** This method is an algorithmic approach of assigning one job to one worker or machine. This method is an efficient solution procedure for assignment problems.

Basic Requirement of Assignment Model

- i. The number of rows must be equal to the number of columns, i.e. square matrix
- ii. The number of lines necessary to cover the zeros must be equal to the number of rows/columns.

Example 1: A manufacturing firm received request to execute three tasks – A, B & C. Based on past records, the firm selected three competent auditors (X, Y & Z), to execute the tasks. Estimated project completion time (in days) is given in the table below:

	A	B	C
X	10	15	9
Y	9	18	5
Z	6	14	3

- i. How should the firm assign the projects so as to minimize the total completion time?
- ii. What is the minimum possible project completion time?

ENUMERATION METHOD

X	Y	Z	Number of days	Completion time
A	B	C	$10 + 18 + 3$	31
A	C	B	$10 + 5 + 14$	29
B	A	C	$15 + 9 + 3$	27
B	C	A	$15 + 5 + 6$	26 ✓
C	A	B	$9 + 9 + 14$	32
C	B	A	$9 + 18 + 6$	33

i) Assignment Strategy:

Auditor X should take Task B

Auditor Y should take Task C

Auditor Z should take Task A

ii) The minimum possible project completion time is 26 days.

HUNGARIAN METHOD

STEPS FOR MINIMIZATION PROBLEMS:

- i. *Row Operation* – select the least element in each row and subtract it from all elements in the row,
- ii. *Column Operation* – select the least element in each column and subtract it from all elements in the column.
- iii. *Optimality Test* – draw lines necessary to cover all zeros in the matrix. If the number of lines equals the number of rows/columns, optimum solution is reached, else embark on Step iv.
- iv. *Improvement* – select the least unlined element, subtract it from all other unlined elements & add it to double lined element(s) (ignoring single lined element)
- v. *Pairing Plan* – concentrate on zeros.

Now we want to apply the above steps to the Example 1. Initial tableau is given below:

	A	B	C
X	10	15	9
Y	9	18	5
Z	6	14	3

Row operation:

	A	B	C
X			
Y			
Z			

Column operation & Optimality Test I:

	A	B	C
X			
Y			
Z			

Since Number of lines $\neq$ Number of rows/columns, optimum solution is NOT yet attained.

Improvement & Optimality test II:

	A	B	C
X	0	0	2
Y	1	5	0
Z	0	3	0

Since Number of lines = Number of rows/columns, optimum solution is attained.

Pairing Plan:

	A	B	C
X	0	0✓	-
Y	-	-	0✓
Z	0✓	-	0

i)

Assignment	Completion time (days)
$X \rightarrow B$	15
$Y \rightarrow C$	5
$Z \rightarrow A$	6
TOTAL	26

ii) Minimum project completion time = $15 + 5 + 6 = 26$ days.

Example 2: O.A.U has decided to build 4 hostels to meet the accommodation problems currently facing the institution. Four contractors (A, B, C and D) are requested to submit bids on each of the hostels. If a contractor is interested in a particular hostel, a very high bid will be submitted for the construction of that hostel. Company A bids for jobs W, X, Y and Z (in ₦ Billion) are 8, 5, 2 and 1 respectively; Company B bids are 5, 8, 6 and 2 respectively; Company C bids are 6, 9, 5 and 3 respectively, while Company D bids are 2, 4, 1 and 5 respectively. The Department of Physical Planning (DPP) of the University, after examining the bids, concludes that there has been no collision and decides to award a job to each contractor. How should DPP assign contractors to jobs in such a way that the total construction cost is minimized? What is the minimum project execution cost?

Solution: Initial tableau

Contractors	Jobs			
	W	X	Y	Z
A	8	5	2	1
B	5	8	6	2
C	6	9	5	3
D	2	4	1	5

Row Operations:

	W	X	Y	Z
A				
B				
C				
D				

Column Operation & Optimality Test I:

	W	X	Y	Z
A				
B				
C				
D				

Improvement & Optimality Test II:

	W	X	Y	Z
A	5	0	0	0
B	1	2	3	0
C	1	2	1	0
D	0	0	0	5

Improvement II & Optimality Test III:

	W	X	Y	Z
A	5	0	0	1
B	0	1	2	0
C	0	1	0	0
D	0	0	0	6

Pairing Plan:

	W	X	Y	Z
A	-	0✓	0	-
B	0	-	-	0✓
C	0	-	0✓	0
D	0✓	0	0	-

Assignment	Completion Cost (₦' Billion)
A → X	5
B → Z	2
C → Y	5
D → W	4
TOTAL	14

OR

Assignment	Completion Cost (₦' Billion)
A → Y	2
B → W	5
C → Z	3
D → X	4
TOTAL	14

Therefore, the minimum project cost = ₦14 Billion

Example 3: A human resource firm employs typists on hourly piece-rate basis for their daily work. There are five typists and their charges and speed are different. According to an earlier understanding only one job is given to one typist and the typist is paid for a full hour even if he works for a fraction of an hour.

Typist	Rate per hour (₦'000)	Number of pages per hour
A	5	12
B	6	14
C	3	8
D	4	10
E	4	11

Job	Number of pages
P	199
Q	175
R	145
S	298
T	178

You are required to:

- present the problem in an assignment model
- assign the jobs to the typists on a one-to-one basis to minimize execution cost
- determine the minimum cost of the project

Solution:

(i) Initial tableau

	P	Q	R	S	T
A					
B					
C					
D					
E					

(ii.) Step1: Row Operation

	P	Q	R	S	T
A					
B					
C					
D					
E					

Step 2: Column Operation & OT

	P	Q	R	S	T
A	2	2	0	4	0
B	6	4	0	10	2
C	0	1	0	1	2
D	2	4	0	4	2
E	2	0	0	0	2

Step 3: Improvement I & OT

	P	Q	R	S	T
A	2	2	2	4	0
B	4	2	0	8	0
C	0	1	2	1	2
D	0	2	0	2	0
E	2	0	2	0	2

Step 4: Improvement II & OT

	P	Q	R	S	T
A					
B					
C					
D					
E					

Step 5: Pairing Plan

	P	Q	R	S	T
A	-	-	-	-	0✓
B	-	-	0✓	-	0
C	0	0✓	-	0	-
D	0✓	-	0	-	0
E	-	0	-	0✓	-

OR

Typist	Job
A	T
B	R
C	Q
D	P
E	S

Typist	Job
A	T
B	R
C	S
D	P
E	Q

iii) Minimum cost of execution = ₦399,000

UNBALANCED ASSIGNMENT PROBLEM

Assignment strategy when number of resources (men, machine, etc.) is not equal to number of tasks. This is often the case in real life situations. Use of dummy to create the balance. The dummy is assigned zero in the optimal assignment solution. Then, follow the same steps either for minimization or maximization problem.

Example 4: A firm received request to execute three jobs – A, B & C. Based on past records, the firm selected four competent hands (P, Q, R, and S) to execute the jobs. Estimated project construction cost (₹ Million) is given in the table below:

	A	B	C
P	10	15	9
Q	9	18	5
R	6	14	3
S	8	16	6

Using Hungarian approach:

- Find the optimal assignment of jobs to the contractors that will minimize the total cost of execution;
- Find for which contractor no job is assigned?
- What is the total cost to complete all the jobs?

Solution: Row Operation:

	A	B	C	Dummy
P				
Q				
R				
S				

Column Operation & Optimality Test I:

	A	B	C	Dummy
P				
Q				
R.				
S				

Improvement & Optimality Test II:

	A	B	C	Dummy
P	-3	0	5	0
Q	2	3	1	0
R	-0	0	0	1
S	1	1	2	0

Improvement II & Optimality Test III:

	A	B	C	Dummy
P	-3	0	5	0
Q	-1	2	0	0
R	-0	0	0	2
S	-0	0	1	0

Pairing Plan:

	A	B	C	Dummy
P	-	0✓	-	0
Q	-	-	0✓	0
R	0✓	0	0	-
S	0	0	-	0✓

(i) Optimal assignment of jobs to the contractors:

Assignment	Construction cost (₦ Million)
P → B	15
Q → C	5
R → A	6
S → Dum	0
TOTAL	26

(ii) No job is assigned to Contractor S

(iii) Minimum construction cost = ₦ 26 Million

IMPORTANCE OF ASSIGNMENT MODEL

The Assignment Model is a valuable tool in the field of Operations Research with several important applications. Here are some key reasons highlighting its importance:

1. Optimal resource allocation: The Assignment Model is used to solve allocation problems where a set of tasks or resources needs to be assigned to a set of individuals or entities. It helps determine the most efficient and optimal assignment of resources to minimize costs, time, or other relevant factors.

2. Workforce management: In staffing and scheduling scenarios, the Assignment Model helps determine the best assignment of employees to specific tasks or shifts. By considering factors like skills, availability, and workload, it assists in optimizing workforce utilization, ensuring balanced workloads, and minimizing idle time.

3. Transportation and logistics optimization: The Assignment Model is applied in transportation and logistics to optimize the allocation of transportation resources, such

as vehicles, drivers, and routes. It helps in minimizing transportation costs, reducing delivery times, and improving overall efficiency in supply chain operations.

4. Project management and task assignment: In project management, the Assignment Model assists in assigning tasks to project team members based on their expertise, availability, and workload. It helps optimize resource allocation, balance workloads, and improve project efficiency by ensuring tasks are appropriately assigned to the most suitable individuals.

5. Facility location and assignment: The Assignment Model aids in determining the optimal assignment of facilities to locations. It considers factors like customer demand, transportation costs, and facility capacities to identify the best possible location for each facility, minimizing overall costs and maximizing service coverage.

6. Healthcare and emergency services: The Assignment Model finds applications in healthcare systems and emergency services to optimize the assignment of medical personnel, equipment, and facilities to maximize response efficiency. It helps in determining the best allocation of resources in emergency situations, reducing response times, and saving lives.

EXERCISE

The table contains information on the cost (₦million) to execute five projects (A, B, C, D and E) by five firms (I, II, III, IV and V). As the production manager, decide how the tasks should be allocated so as to minimize the total cost of execution? What is the total minimum cost of execution?

	A	B	C	D	E
I	4	5	9	8	7
II	6	4	8	3	5
III	7	3	10	4	6
IV	5	2	5	5	8
V	6	5	3	4	9